

OF GOOGLE GLASS AND HMDs



Maybe you have heard of Google Glass? Glass is an amalgamation of technologies old and new, and one that Google hope will usher in an age of practical head-mounted-displays (HMD), augmented reality (AR), and a brand new revenue stream. The Glass fuses a 1.3 cm display into the frame of a pair of glasses. The precise science behind Glass is currently unknown but one will be able to see a phone-like interface, powered by Android, through the 1.3cm window. The interface will allow the user to take photos, take part in video conferences (Google+

Hangout), check appointments, and access the Web and Google Maps. Google isn't the only player looking to make an impact in the HMD business. Apple too is in the game, or at least thinking about the game: it won a patent late in 2012 for 'peripheral treatment for head-mounted displays'. Apple's implementation is binocular and sends two different images to each eye: "Apple hopes to solve many longstanding issues with HMDs, particularly the general risk of eyestrain that could be caused by the difference in distances between the wearer's field of vision and the peripheral display itself. Essentially, Apple's technology dynamically matches the color images being transmitted with either LED, OLED or lasers, which are converged stereoscopically to reduce the tunnel effect experienced in many of today's HMDs".

Sony has experimented with HMDs as well, somewhat successful too: their HMZ-T1 'personal 3D viewer' consisted of two OLED screens and offers the user a 720p display for each eye. The HMZ simulates the viewing experience of a 700-inch screen, as viewed from a distance of 20 meters... Recently, the Oculus Rift has been making the waves in the gadget circles: winning quite a few hearts first and raising a cool \$2.4 million through Kickstarter, followed up by winning quite a few best-of-show awards at CES 2013 (especially noted for its superior head-tracking). The Rift

is an HMD endorsed by the likes of John Carmack and Gabe Newell – it will offer an effective resolution of 640x800 to each eye in stereoscopic 3D. The Oculus will first be offered as a developer version this year, followed by a consumer version, probably in early 2014, or late 2013. The consumer version will have a higher resolution and better quality lenses.



Green (short for Mr Green Dot the courier boy) and asked Odin to chart out his whereabouts for the next 30 minutes. Thirty seconds later, my HUD filled with five routes, with varying thickness – the thicker the route, the more likely Green was to take it – it was a projection based on his past routes at this time of day, taking into account the route he had taken in the past fifteen

minutes. A routine tracing algorithm. I had thirty minutes to get to Holy Family High School, which was to be the courier's destination, a certainty rank of 95%. I grabbed the box of grandpa-pants, pinged my car through Odin (it would drive itself out to the street from the parking lot), and headed for the door. Sensing my exit, OdinVision reminded me that I hadn't bathed in three

days now – perhaps a hot shower first? That didn't slow me down; it only darkened my mood further. This led to two things: the blues music was now routed directly to my personal area network to continue an attempt to better my mood, and the door sprayed me with a deodorant and my favourite perfume as I head out, a futile attempt to drown my manly, err, musk.

The car was waiting for me. As I approached, it detected my presence, pinged my personal area network for an authentication code and scanned my face as a double-confirmation. Satisfied, it opened the door. I threw in the box and took to the wheels. The car could self-drive but cursing at the traffic and narrowly avoiding the pedestrians was a hobby, it would improve my mood by a degree or two. As I touched the steering, the car scanned my fingerprints, then started its electric engine, threw up my destination and charted out the shortest route based on traffic conditions. I chose a slightly denser road; I had people to narrowly miss, after all...

Mumbai in 2050 is much like Mumbai in 20-whatever. It always has been and always will be crowded, smelly, noisy and very very busy. Too busy to notice you, and be thankful for that... some things have improved though: the electric cars have dialed down the poisoned air by a few notches, the billboards are all mostly relegated to the virtual – the likes of OdinVision now routinely fill your peripheral vision with advertisements – personalised, tailored ads, honed for maximum ROI. And there are no cops in sight. This last part is not because crime is low, or because Mumbaikars have suddenly grown road sense, the reason is more practical: since every transaction is now electronic – done over encrypted networks using bits and blobs of software – the scope for bribes has greatly reduced. And thus so has the cops' main incentive to ply the streets for the next victim...

I narrowly avoided: three school children, one granny, two kittens, one biker, and one extremely inebriated soul who seemed smitten by my car. Odin awarded me 15,000 points at this feat and unlocked the Road Warrior achievement. Having reached my destination fifteen minutes ahead of time, I headed to the KFC next door for a quick brunch. Just a few feet from the KFC door, I bumped into someone who